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TITLE

Web-Based Interactive CCU Orientation Manual: Eliminating Arterial Line-Related Compartment Syndrome Through Digital Education

(18 words)

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ABSTRACT

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BACKGROUND

Arterial line placement is a routine procedure in coronary care units (CCU), yet poses significant bleeding risks for acute myocardial infarction (AMI) patients receiving dual antiplatelet therapy (DAPT) combined with anticoagulation. Between 2022–2025, our 10-bed CCU experienced five cases of brachial artery puncture-related compartment syndrome requiring emergency fasciotomy. Root cause analysis using the Human Factors Analysis and Classification System (HFACS) identified critical gaps: absence of standardized arterial line placement guidelines, inadequate risk stratification for anticoagulated patients, and insufficient education on DAPT bleeding complications.

SUMMARY OF WORK

We developed a comprehensive web-based CCU orientation manual as an educational intervention. This React-based single-page application features nine interactive modules: daily checklists, safety case studies (including the compartment syndrome cases), ACS medication orders, heart failure certification workflows, special post-procedure care protocols, drug reference tables, emergency algorithms, chart writing guides, and self-assessment quizzes. The intervention specifically embedded: (1) new arterial line placement guidelines prioritizing radial artery with ultrasound guidance, (2) explicit prohibition of brachial artery puncture in patients on Brilinta/Prasugrel, and (3) standardized 6P monitoring protocol (Pain, Paralysis, Pallor, Polar, Paresthesia, Pulseless) post-procedure. The manual was deployed in September 2025 as mandatory orientation material for all CCU residents and nurse practitioners.

SUMMARY OF RESULTS

Following implementation, we achieved zero arterial line-related compartment syndrome cases over six months of monitoring (August 2025–January 2026), representing a **100% reduction** from the previous incidence of 1.6 cases/year. Secondary outcomes demonstrated improvement: radial artery first-choice rate increased from 65% to 95%, ultrasound-guided puncture rate rose from 40% to 90%, 6P monitoring protocol compliance reached 100%, and 92% of learners achieved passing scores ($\geq$80%) on the integrated knowledge assessment quiz.

DISCUSSION AND CONCLUSIONS

This quality improvement initiative demonstrates that HFACS-driven root cause analysis, combined with web-based interactive educational tools, can effectively eliminate preventable procedural complications. The just-in-time accessibility, interactive case-based learning, and integrated self-assessment features of digital education platforms offer advantages over traditional orientation methods.

TAKE-HOME MESSAGES

1. Web-based interactive education effectively reduces high-risk procedural complications
2. HFACS framework enables systematic identification of educational gaps in patient safety
3. Digital orientation manuals provide accessible, updateable just-in-time learning resources

Keywords: Patient safety, Digital education, Compartment syndrome, HFACS, Quality improvement, Coronary care unit, Procedural complications

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